Cocktail AI



Generated by Google Gemini

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THE PROJECT:

Unsupervised machine learning model that uses:
- TF-IDF, Cosine Similarity, Clustering,

A user is prompted for a list of ingredients for a cocktail.

The model returns:
- Its top 5 recommended drinks based on those ingredients + cluster(s) they're contained in
- Similarity score of inputted ingredients to the cocktail's
- An option to view further information (ingredients, directions, etc.)

```
Your top 5 cocktail recommendations:
        1. Sweet Bananas - Similarity: 58.41% (Cluster #3)
        2. Bahama Mama - Similarity: 51.42% (Cluster #3)
        3. Fruit Shake - Similarity: 49.41% (Cluster #3)
        4. Bubble Gum - Similarity: 48.20% (Cluster #3)
        5. Zombie - Similarity: 22.56% (Cluster #3)
```



From: www.thespruceeats.com

DATASET

424 unique drinks
- Attributes included (10):
  - Serial #, Id, name, alcoholic, category, glass type, instructions, drink thumbnail, ingredients, measurements

Processing:
- Dropped serial #, thumbnail
- Binary encoded alcoholic (1) & non alcoholic (0)

```
id  name  alcoholic        category       glassType
 0    A1          1        Cocktail  Cocktail glass
 1   ABC          1            Shot      Shot glass
 2   Ace          1        Cocktail   Martini Glass
 3  ACID          1            Shot      Shot glass
 4  Adam          1  Ordinary Drink  Cocktail glass

                                          instructions
Pour all ingredients into a cocktail shaker, m...
                            Layered in a shot glass.
Shake all the ingredients in a cocktail shaker...
Poor in the 151 first followed by the 101 serv...
In a shaker half-filled with ice cubes, combin...

                                          ingredients
['Gin', 'Grand Marnier', 'Lemon Juice', 'Grena...
    ['Amaretto', 'Baileys irish cream', 'Cognac']
['Gin', 'Grenadine', 'Heavy cream', 'Milk', 'E...
                  ['151 proof rum', 'Wild Turkey']
         ['Dark rum', 'Lemon juice', 'Grenadine']

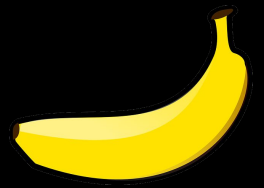
                                   ingredientMeasures
['1 3/4 shot ', '1 Shot ', '1/4 Shot', '1/8 Sh...
                           ['1/3 ', '1/3 ', '1/3 ']
['2 shots ', '1/2 shot ', '1/2 shot ', '1/2 sh...
                       ['1 oz Bacardi ', '1 oz ']
                     ['2 oz ', '1 oz ', '1 tsp ']
```

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TF-IDF

Term Frequency - Inverse Document Frequency (TF-IDF)

- TF: measure of how important a single ingredient is in **A SINGLE** cocktail
- IDF: measure of how important (uniqueness) a single ingredient is across **ALL** cocktails
- TF-IDF (product of the two)
  - Result is the importance of an ingredient in a cocktail while still considering how common it is in all cocktails
- EX: Cocktail gives a high TF-IDF for “Bananas” if:
  - The cocktail mentions bananas often (TF)
  - Banana is not common in other cocktails (IDF)
- Each ingredient in each cocktail gets a TF-IDF score
- Cocktail becomes a vector of these scores



.32



.00



.12

Implementation

Cosine Similarity

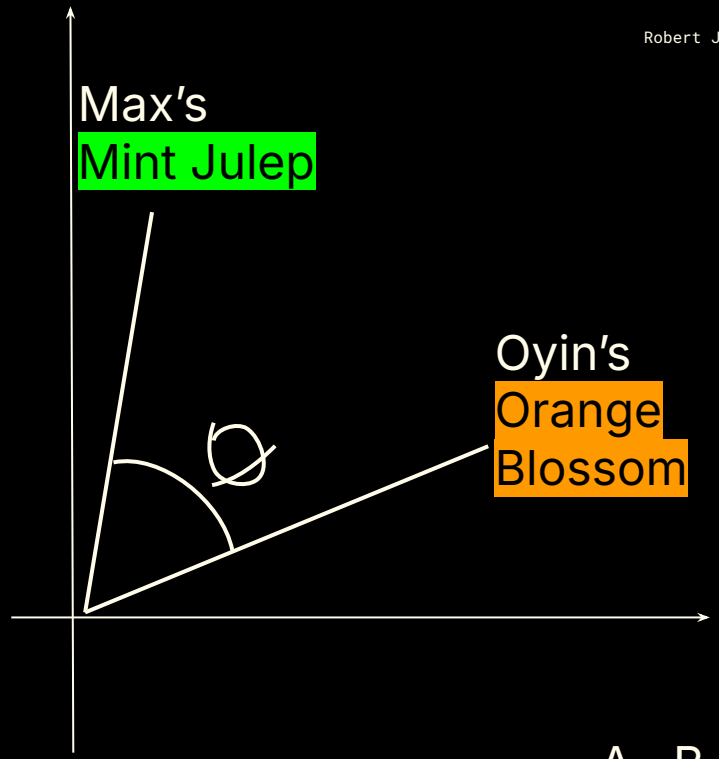
At this point, cocktails are TF-IDF vectors (list of numbers)

- If you have two cocktails, how do you measure how similar they are?

Cosine Similarity

This method measures the angle between the two vectors

- 0 <= value <= 1
- Higher the score, greater the similarity



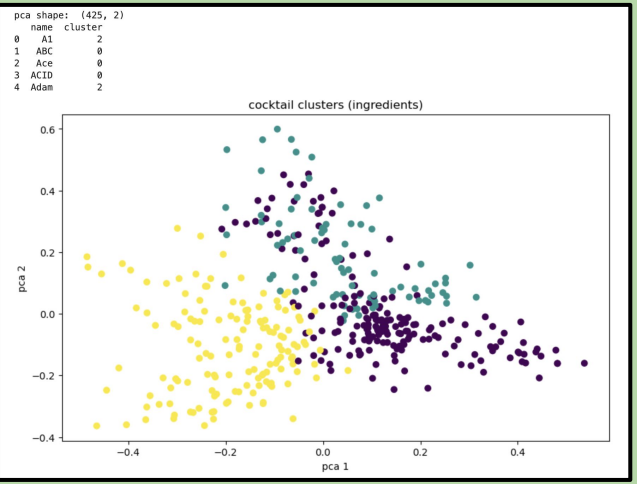
$$\text{Cos_sim}(A, B) = \cos(\Theta) = \frac{A \cdot B}{|A|\,|B|}$$

Putting it all together: Clustering

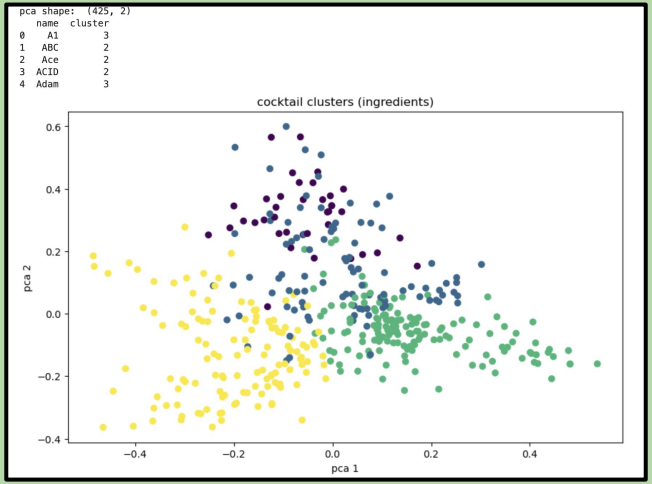
The clustering process uses the cosine similarity scores to group cocktails together.

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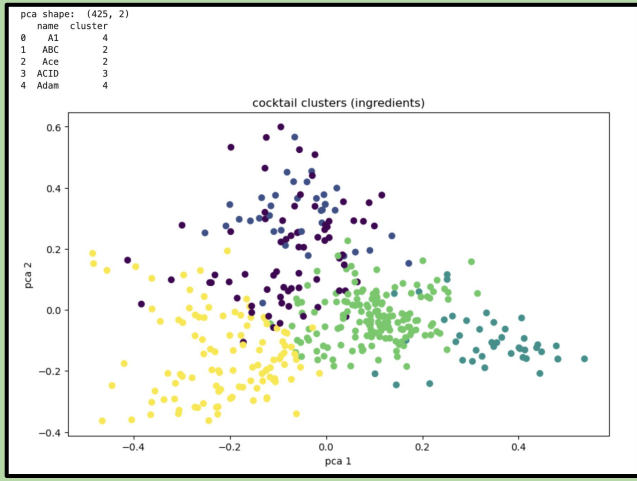
RESULTS



3 Clusters



4 Clusters



5 Clusters

RESULTS PT.2 (3 clusters)

```
************************************************************
Enter a list of ingredients, separated by commas and a space. For example: rum, lime jui
ce, sugar. Or, type 'done' to exit:  Kahlua, Baileys irish cream, Frangelico, Coffee, Cr
eam

Your top 5 cocktail recommendations:
        1. Afternoon - Similarity: 100.00% (Cluster #0)
        2. Nutty Irishman - Similarity: 77.35% (Cluster #0)
        3. Dirty Nipple - Similarity: 66.36% (Cluster #0)
        4. Bumble Bee - Similarity: 66.36% (Cluster #0)
        5. Hot Creamy Bush - Similarity: 64.55% (Cluster #0)
```

Afternoon (control):
Kahlua, Baileys irish cream, Frangelico, Coffee, Cream
Nutty Irishman:
Baileys irish cream, Frangelico, Milk
Dirty Nipple:
Kahlua, Sambuca, Baileys irish cream

```
Enter a list of ingredients, separated by commas and a space. For example: rum, lime jui
ce, sugar. Or, type 'done' to exit:  orange juice, banana, vodka, mint

Your top 5 cocktail recommendations:
        1. Bubble Gum - Similarity: 58.71% (Cluster #2)
        2. Screwdriver - Similarity: 53.55% (Cluster #2)
        3. Fruit Shake - Similarity: 50.54% (Cluster #2)
        4. Sweet Bananas - Similarity: 42.91% (Cluster #2)
        5. Bahama Mama - Similarity: 39.19% (Cluster #2)
```

Bubblegum
Vodka, Banana liqueur, Orange juice, Peach schnapps
Screwdriver
Vodka, Orange Juice
Fruit Shake
Yogurt, Banana, Orange Juice, Fruit, Ice

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BONUS (output)



www.wearenotmartha.com

```
If you like a recommendation, type and enter its corresponding number (1-5). 'new' to en
ter a new list.
 Your choice:  1

************************************************************

************************************************************

Drink: Bubble Gum

************************************************************

Category: Shot

************************************************************

Glass Type: Shot glass

************************************************************

Ingredients:
1/4 of Vodka
1/4 of Banana liqueur
1/4 of Orange juice
1/4 of Peach schnapps

************************************************************

Instructions: Layer in order into a shot glass.

************************************************************

If finished, type 'done'

If you like a recommendation, type and enter its corresponding number (1-5). 'new' to en
ter a new list.
 Your choice:  done

************************************************************
Enjoy the cocktail!
```